

1X10

~~d) INTR~~

(x) If the microprocessor is capable of addressing 1MB memory, its address bus width is

- a) 16 bits
- b) 20 bits
- c) 8 bits
- d) None of these

(xi) Instruction queue of 8086 consists of

- a) 4 data
- b) 6 data
- c) 8 data
- d) 16 data

(xii) The data bus of 8085 microprocessor is

- a) Unidirectional
- b) Bidirectional
- c) Unidirectional as well as Bidirectional
- d) None of these

GROUP B
(Long answer type questions)
Answer any seven

(5+5)

2 (a) Explain the need to demultiplexing the bus AD_0-AD_7 . Discuss the function of ALE and HOLD signals of 8085 microprocessor.

(b) What are the flags in 8085? Discuss the flag register with some examples.

3 (a) Explain the need of program counter and stack pointer. Why are PC and SP 16-bit register?

(b) Differentiate between memory mapped I/O and peripheral mapped I/O.

(5+5)

4 Write a program to separate the odd and even numbers in a series of N data bytes. The series starts from 8200H and the value of N is available at 8300H. Store the odd numbers from 8400H and the even numbers from 8500H.

(10)

5 (a) PUSH and POP are the only instructions of 8085 that uses stack. Comment on this statement. Write a program to clear the flags of 8085.

(6+4)

(b) Draw the schematic diagram to generate Read/Write control signals for memory and I/O devices in the 8085 microprocessor.

6 Interface the 16KB RAM, 8 KB EPROM and 4 KB EPROM memory devices in 8085 microprocessor in absolute decoding method.

(10)

7 (a) Explain the role of stack in executing a subroutine.

(4+6)

(b) Draw the timing diagram for the instruction XRA M stored in memory location 9550H.

8 (a) Draw and explain the control word format of 8255. Write a control word for port A and port C as an output port and port B as an input port.

(5+5)

(b) Draw and discuss the functional block diagram of 8255.

9 (a) Draw the architecture of 8086 microprocessor. What are the main functions of BIU and EU of 8086 microprocessor.

(7+5)

(b) Explain how 20-bit physical address is generated in 8086 microprocessor?

10 Write a program to convert a hexadecimal number stored in memory location 8200H to its corresponding ASCII. Store the result from 8300H.

(10)

11 (a) How does an 8085 microprocessor responded to INTR interrupt signal?

(5+5)

(b) What do you mean by hardware and software interrupts of 8085? Write a program to enable RST 7.5, RST 6.5 and disable RST 5.5.